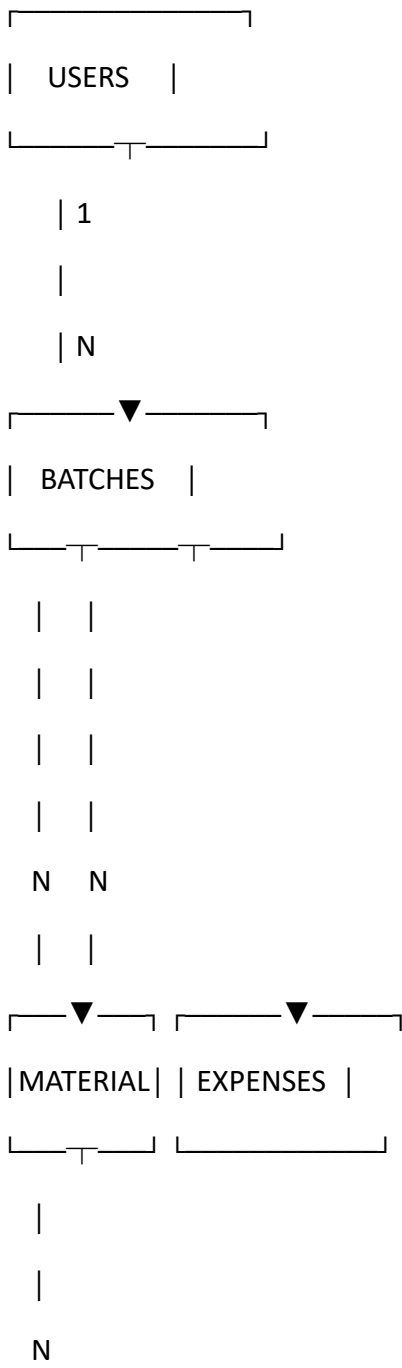
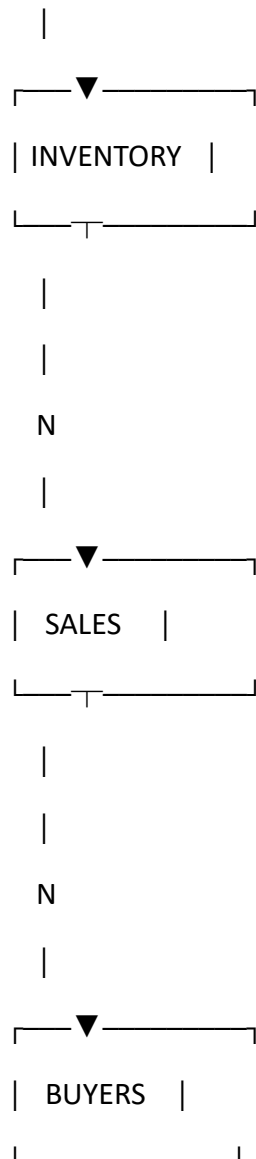


The next artifact should be **ERD (Entity Relationship Design)**, because before writing SQL tables or FastAPI models, you need a final relationship map.

ERD – ScrapFlow AI v1.0

High-Level ERD





Relationship Details

User → Batch

1 User



Many Batches

Example:

Zaid

Batch-001

Batch-002

Batch-003

Batch → Materials

1 Batch



Many Materials

Example:

Batch-001

Copper

Steel

Plastic

Batch → Expenses

1 Batch



Many Expenses

Example:

Batch-001

Fuel

Maintenance

Transport

Material → Inventory

Current Design:

Material



Inventory Update

Inventory acts as an aggregated view.

Example:

Copper Batch 1 = 20kg

Copper Batch 2 = 50kg

Inventory Copper = 70kg

Buyer → Sales

1 Buyer



Many Sales

Example:

ABC Metals

Sale 1

Sale 2

Sale 3

Material → Sales

1 Material



Many Sales

Example:

Copper

Sale 1

Sale 2

Recommended Improved Database

I would slightly improve the current DDD.

Instead of:

materials

Use:

batch_materials

Reason:

More descriptive.

Inventory Table Improvement

Current:

inventory

id

material_type

available_weight

Recommended:

inventory

id

material_type

available_weight

average_cost

estimated_value

updated_at

Why?

Future dashboard becomes easier.

Sales Table Improvement

Add:

sale_status

Values:

Completed

Pending

Cancelled

Useful for future credit sales.

Expense Table Improvement

Add:

expense_date

Instead of relying only on created_at.

Future AI Tables

Don't create now.

But reserve:

voice_notes

ai_queries

notifications

for V2.

Final Database Modules

Core Tables

Users

Batches

Batch_Materials

Inventory

Buyers

Sales

Expenses

Future Tables

Voice_Notes

AI_Queries

Notifications

After ERD

The next deliverable should be:

Wireframe Specification Document (WSD)

This is different from UXSD.

UXSD says:

What screens exist

WSD shows:

Exactly what appears on each screen

Example:

Dashboard

Today's Profit

₹12,500

Inventory Value

₹2,15,000

Recent Sales

...

Recommended Final Documentation Order

01 Product Blueprint 

02 DDD 

03 BASD 

04 BDFS 

05 UXSD 

06 ERD  (Current)

07 WSD (Next)

08 SQL Schema

09 FastAPI Project Setup

10 Development Sprint Plan

11 Development

At this point you're about **95% done with planning**. After the Wireframe Specification Document, I would stop creating documents and begin building the database schema and FastAPI backend.